

Persona 1: Conflicted Power User (Student)

1. User / Protagonist : Student

Description:

18–22-year-old undergraduate student who frequently uses AI tools for assignments, studying, and productivity. Highly tech-comfortable, relies on AI under time pressure, prioritizes convenience, but lacks clear understanding of AI limitations and risks.

2. Context & Situation

When/Where:

- During assignments, exams, project work
- Especially under deadlines or time pressure
- In academic environments with increasing AI usage

Trigger:

- Need to complete work quickly
- Difficulty understanding topics
- Peer influence (others using AI)

Environment:

- Digital learning + personal devices
- Minimal supervision while using AI

Secondary Stakeholders:

- Professors / institutions
- Employers (long-term skill impact)
- AI platforms

3. Evidence Base

Methods Used:

- Semi-structured interviews
- Conversational (open-ended) field interviews
- Persona synthesis

Key Data Points:

- “I use it almost daily”
- “If it’s important, I cross-check... otherwise I just use it”
- “I don’t know where the line is between help and cheating”
- “It saves time but I’m not sure if it’s always correct”

Observation: Users continue using AI **despite uncertainty**

4. Pain Points & Needs

Frustrations:

- Cannot judge when to trust AI
- Lack of awareness of limitations
- Inconsistent verification behavior
- Over-reliance without understanding

Unmet Needs:

- Ability to make **better decisions while using AI**
- Awareness of risks and limitations
- Confidence in when/how to rely on AI

5. Root Causes (5 Whys)

General Causes

- Users rely on AI due to convenience
- They lack awareness of limitations and risks
- AI systems provide outputs without contextual guidance
- Systems are designed for performance, not user understanding
- Transparency and interpretability are limited in AI design

Deep Root Cause:

Lack of **trust calibration and awareness infrastructure** — users are not equipped to judge when to trust, question, or verify AI outputs.

6. Impact & Scale

Severity: High

- Affects learning quality and decision-making

Frequency: Very high

- Daily/weekly usage among students

Scale:

- Majority of students using AI tools

Consequences if unresolved:

- Misplaced trust in incorrect outputs
- Reduced critical thinking
- Over-dependence on AI
- Poor decision-making

Current Workarounds:

- Occasional cross-checking
- Editing AI responses
- Avoiding AI in important cases

Why they fail:

- Inconsistent
- Based on guesswork
- Do not build true understanding

7. Point of View (POV Statement)

Students who frequently use AI need to make better decisions about when and how to rely on it because they lack awareness of its limitations and cannot accurately judge its trustworthiness.

8. How Might We (HMW)

- How might we help students decide when to trust AI outputs?
- How might we make users more aware of AI limitations without disrupting convenience?
- How might we reduce over-reliance while still preserving productivity?
- How might we help users verify AI responses more effectively?
- How might we design AI interfaces that visually communicate certainty levels or source citations, especially for complex technical tasks like debugging or system architecture?
- How might we create "productive friction" in the AI experience that slows users down just enough to engage their critical thinking before copying text?
- How might we prompt students to reflect on and summarize the AI's logic in their own words to ensure comprehension?

- Hmw introduce small moments of reflections before using ai blindly
- Shift from passive user to active learner
- How might we shift the user's perception of AI from an infallible "answer engine" to a collaborative "sounding board"?
- How might we seamlessly integrate rapid, independent fact-checking tools directly into the student's existing digital learning environment?
- How might we help students track their own independent learning progress separately from the AI's generated outputs?
- How might we reward users for independently verifying and correcting AI-generated claims?

9. Reframe Check

Alternative framing:

- What if the problem is not misuse, but **misplaced trust**?
- What if users don't need rules, but **better judgment tools**?

Who might see this as a feature?

- Power users who benefit from speed over accuracy
- Companies optimizing for engagement and usage

10. Final Problem Statement

Conflicted student users experience misplaced trust, inconsistent verification, and over-reliance on AI because they lack awareness of its limitations and the ability to judge its reliability.

How might we help them make better decisions while using AI so that they can maintain productivity while improving accuracy, confidence, and responsible usage?

Persona 2: Cautious Professional

1. User / Protagonist : Working Professional

Description:

23–35-year-old working professionals who use AI tools regularly for productivity, automation, and efficiency. Highly tech-comfortable, but operate in environments with accountability, accuracy requirements, and data sensitivity. They value efficiency but cannot afford errors.

2. Context & Situation

When/Where:

During professional tasks such as coding, drafting emails, analysis, reporting, and decision-making

Trigger:

- Need to complete tasks efficiently
- Repetitive or time-consuming work
- Pressure to maintain output quality

Environment:

Professional workplace with accountability and performance expectations

Secondary Stakeholders:

- Employers / organizations
- Clients
- Team members
- AI platforms

3. Evidence Base

Methods Used:

Semi-structured interviews

Conversational field interviews

Persona synthesis

Key Data Points:

- “I never trust it blindly”
- “I always cross-check outputs”
- “It depends on the context”
- Observation: Users verify AI outputs before use

4. Pain Points & Needs

Frustrations:

- AI produces confident but incorrect outputs
- Lack of transparency in reasoning
- Time spent verifying reduces efficiency
- Concerns about privacy and misuse

Unmet Needs:

- Faster and reliable verification
- Trustworthy outputs
- Safe integration into workflows

Emotional Impact:

- Skeptical
- Mentally fatigued
- Slightly frustrate

5. Root Causes (5 Whys)

AI is used for efficiency

- Outputs require verification
- Outputs are not fully reliable
- AI lacks contextual understanding and validation
- Systems prioritize generation over correctness

Deep Root Cause:

Lack of reliability and verification infrastructure — AI systems are not designed to meet professional standards of accuracy and accountability

6. Impact & Scale**Severity:** High

Affects work quality and productivity

Frequency: High

Daily usage in workflows

Scale:

Large number of professionals across industries

Consequences if unresolved:

- Reduced trust in AI
- Increased workload (verification burden)
- Risk of errors in professional outputs

Current Workarounds:

- Manual checking
- Testing outputs

- Limited AI usage

Why they fail:

- Time-consuming
- Reduces efficiency benefits
- Still prone to human oversight errors

7. Point of View (POV Statement)

Working professionals who use AI tools need to efficiently verify and trust AI-generated outputs because they are fully accountable for the results and cannot rely on systems that produce confident but potentially incorrect information.

8. How Might We (HMW)

- How might we help professionals quickly verify AI outputs?
- How might we reduce the burden of manual validation?
- How might we help professionals trust the outputs?
- How might we make AI reasoning more transparent?
- How might we ensure reliable outputs in professional contexts?
- How might we integrate automated validation into AI workflows?
- How might we let professionals set accuracy levels based on how important the task is?
- How might we add confidence scores to each claim in an AI response?
- How might we let professionals flag risky outputs so the AI learns what to double-check next time?
- How might we use a second AI to double-check the first one's work?
- How might we clearly show when a task is safe to run through AI without breaking privacy rules?

9. Reframe Check

Alternative framing:

What if the problem is not AI inaccuracy, but lack of built-in validation systems?

Who might see this as a feature?

Companies optimizing for speed over reliability

10. Final Problem Statement

Cautious professional users experience skepticism, verification fatigue, and reduced efficiency when using AI because outputs are not inherently reliable and require manual validation.

How might we help them trust and verify AI-generated outputs efficiently so that they can maintain productivity without compromising accuracy, accountability, and data security?

Persona 3: High-Stakes Skeptic

1. User / Protagonist : High stakes professional

Description:

25–50-year-old professionals working in high-risk environments such as healthcare and education. Moderate tech comfort. They prioritize accuracy, ethics, and safety over speed and convenience.

2. Context & Situation

When/Where:

During critical decision-making tasks involving patient care, teaching, or evaluation

Trigger:

- Need for high accuracy
- Risk-sensitive situations
- Ethical responsibility

Environment:

High-stakes, regulated, and accountability-driven environments

Secondary Stakeholders:

- Patients / students
- Institutions
- Regulatory bodies
- Society

3. Evidence Base

Methods Used:

Interviews

Observational insights

Key Data Points:

- “Even a small mistake can mislead”
- “I always check everything”
- “AI should not replace judgment”

4. Pain Points & Needs

Frustrations:

- AI is unreliable in critical contexts
- No clear safe vs unsafe usage boundary
- Ethical concerns about misuse

Unmet Needs:

- Safe and reliable AI
- Clear boundaries of usage
- Domain-specific trust

Emotional Impact:

- Cautious
- Risk-aware
- Responsible

5. Root Causes (5 Whys)

AI is avoided in critical tasks

- Risk of incorrect outputs
- No guarantees of reliability
- AI is general-purpose
- Not designed for high-stakes environments

Deep Root Cause:

Lack of domain-specific safety, accountability, and boundary systems in AI design

6. Impact & Scale

Severity: Very High

Errors can cause serious harm

Frequency: Moderate but critical

Scale:

Healthcare, education, law, and other high-risk domains

Consequences if unresolved:

- Avoidance of AI
- Missed efficiency benefits
- Risk of misuse by others

Current Workarounds:

- Avoid AI in core tasks
- Manual verification
- Limited use

Why they fail:

- Prevents meaningful adoption
- Limits benefits of AI

7. Point of View (POV Statement)

High-stakes professionals need to use AI safely within clearly defined boundaries because errors in AI outputs can have serious consequences and current systems do not provide sufficient reliability or safeguards.

8. How Might We (HMW)

- How might we define safe boundaries for AI usage?
- How might we communicate risk and uncertainty clearly?
- How might we design domain-specific safeguards?
- How might we prevent AI from overstepping critical decisions?
- How might we use AI as a second opinion instead of the main decision-maker?
- How might we design AI that explains its limits before answering, not after?
- How might we keep a record of AI's role in every critical decision?
- How might we require expert review before AI outputs are used in high-stakes cases?
- How might we let AI flag ethical concerns in a decision without making the decision itself?
- How might we involve patients, students, or other affected people in understanding how AI was used?

9. Reframe Check

Alternative framing:

What if the problem is not trust, but misalignment between AI design and high-stakes environments?

Who might see this as a feature?

General-purpose AI systems optimized for broad usability

10. Final Problem Statement

High-stakes professionals experience limited adoption and low trust in AI because current systems are not designed for reliability, safety, and accountability in critical environments.

How might we design AI tools that provide clear boundaries, reliability, and safeguards so that they can be used confidently without risking safety, ethics, or professional responsibility?

Persona 4: Low-Trust Casual User

1. User / Protagonist : Low Trust Casual User

Description:

30–55-year-old occasional AI users with moderate to low tech comfort. Use AI for simple tasks but avoid relying on it for important decisions due to lack of trust.

2. Context & Situation

When/Where:

During everyday tasks like searching information, understanding concepts, or drafting simple content

Trigger:

- Need for quick answers
- Convenience

Environment:

Low-pressure personal usage

Secondary Stakeholders:

- Family
- Service providers

- General public

3. Evidence Base

Methods Used:

Interviews

Key Data Points:

- “I usually cross-check”
- “I’m not sure if it’s correct”
- Uses AI only for simple tasks

4. Pain Points & Needs

Frustrations:

- Cannot trust AI answers
- Lack of sources or proof
- Confusing outputs

Unmet Needs:

- Clear trust indicators
- Easy verification
- Confidence in usage

Emotional Impact:

- Uncertain
- Hesitant
- Confused

5. Root Causes (5 Whys)

Users do not trust AI

- Cannot verify answers
- No sources or signals
- Lack of transparency
- Black-box system design

Deep Root Cause:

Lack of transparency and trust communication in AI systems

6. Impact & Scale

Severity: Moderate

Frequency: Occasional

Scale: Large general population

Consequences if unresolved:

- Underuse of AI
- Inefficiency
- Reliance on traditional methods

Current Workarounds:

- Googling
- Asking others
- Avoiding AI

Why they fail:

- Adds effort
- Reduces convenience

7. Point of View (POV Statement)

Casual AI users need to understand whether AI outputs are trustworthy because current systems do not provide clear signals of accuracy, sources, or confidence.

8. How Might We (HMW)

- How might we make AI answers easier to verify quickly?
- How might we show clear sources and evidence?
- How might we signal uncertainty transparently?
- How might we reduce the need for cross-checking?
- How might we let users rate answers so they build their own sense of what AI gets right?
- How might we make AI admit "I don't know" in a friendly way instead of guessing?
- How might we use everyday comparisons to explain how much AI can be trusted?
- How might we warn users when AI isn't reliable for certain topics (medical, legal, etc.)?
- How might we let users compare AI's answer with a quick human-written one side by side?
- How might we help casual users build simple habits for spotting wrong answers?

9. Reframe Check

Alternative framing:

What if the issue is not trust, but lack of visible trust signals?

Who might see this as a feature?

Systems designed for simplicity over transparency

10. Final Problem Statement

Low-trust casual users experience uncertainty and limited reliance on AI because they cannot easily judge the trustworthiness of its outputs.

How might we help them quickly assess reliability so they can confidently use AI without needing constant manual verification?